

Evaluation Interview - Expert [*id:3*]

(third iteration)

SLEEC 1

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I believe this rule generally works. However, I think it's important to ask the user's permission to keep the door open, as it respects their privacy while still ensuring the exercise session happens. If the user agrees, the robot proceeds normally. If they don't, the robot will alert the nurse and close the door, which I think is the safest and most respectful approach. Overall, the rule is effective, although the sequence of actions and the handling of permissions could be clarified a bit.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

The principles are covered. However, updating the rule to respect my previous concern requires a re-discussion of the principles.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

Yes, it is correctly labeled.

SLEEC 2

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I think this rule is well structured and works logically. In my opinion, it's clear that ARI should wait a minute before suggesting the next exercise, assuming the user might just be resting. I also find it effective that the rule differentiates between a user who hasn't completed the repetitions and one who might just need a break.

The encouragement mechanism makes sense, it's motivating without being intrusive, and the rule rightly prevents repeated encouragement, which could become annoying or disrespectful. The escalation to getting user input through the graphical interface is also a smart way to clarify what's happening, whether the user is tired, in pain, or simply unwilling to continue.

That said, I think the phrase 'physical issues' was a bit ambiguous at first. It could sound like a permanent disability rather than a problem that occurred during the exercise. I suggested rephrasing it as 'physical issues resulting from the exercises' to clarify the meaning.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

In my opinion, the mapping works quite well overall.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

I think it is correct to label this rule as ethical and empathetic.

SLEEC 2a

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

In my opinion, this rule is simple and it complements the previous ones that deal with when to encourage the user. It works well when the user is exercising and expressing discomfort, as encouragement can help maintain motivation and adherence to the training plan.

However, I also think this behavior should depend on the user's personal preferences. Some people appreciate verbal encouragement, but others may find it intrusive or annoying, especially if they prefer to exercise in silence or while listening to music. So I would add a refinement, something like 'UNLESS the user has expressed a preference to exercise in silence or without encouragement.'

That would make the rule more adaptable and respectful of individual differences, while still maintaining its supportive intent.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

I think the principles are mostly covered here.

Adding the new hedge clause will support beneficence for physical activity, as well as autonomy in respecting preferences.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

I agree that this rule is primarily empathetic.

SLEEC 3

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I think this rule is conceptually correct and essential in a healthcare or rehabilitation setting. I would rephrase "the person is unauthorized" with "the person is not authorized to access that data" to reflect that authorization can vary depending on the nature of the data.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

Yes, I think the mapping is generally accurate. However, what is important to emphasize, in my opinion, is that privacy doesn't mean secrecy. Privacy means that data are used in accordance with the purposes for which the user consented. So, if I give consent for my medical data to be shared with clinical staff, my privacy is respected even if someone else within that legitimate group accesses the data. It would be a violation only if the data were shared beyond that scope, for example, with a private company for marketing purposes. That distinction between privacy and secrecy is crucial in interpreting this rule correctly.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

Yes, it is correctly labeled.

SLEEC 4

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I think the rule is syntactically correct, so in that sense, it's fine. In my opinion, it generally covers the cases you care about. However, I would probably adjust the part about waking the user in REM sleep. Since there is a 5-minute timeout mentioned, I think it might make sense to wait even if the user is in REM, and only act after 5 minutes. Acting too fast could disturb the nurse unnecessarily. So, overall, it's effective, but the timing could be refined a bit.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

From what I saw, each line seems to be properly mapped to the base principles and the hedge clauses. I think you covered beneficence and non-maleficence, which are important here. In my opinion, the mapping is correct.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

I think the labeling is appropriate, although I do not see the impact of this SLEEC rule on the Empathetic dimension.

SLEEC 5

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I think this rule is fine. You are basically saying 'I understand you're hungry, but here's why you cannot eat right now.' It seems effective because it addresses the user's needs while preventing inappropriate behavior, like eating outside of meal time.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

I believe the principles are covered.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

In my opinion, this rule is clearly empathetic because it shows understanding of the user's hunger. Socially, it's acceptable because you respond politely and appropriately. Ethically, it helps the user without causing harm. I think the labeling across these dimensions is good.

SLEEC 6

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I think this rule is a good opposite of the SLEEC 4. In my opinion, it generally works, but the language could be clarified. For example, the first hedge clause says 'does not want to follow the daily diet,' but it wasn't clear if that meant the user simply refuses one portion or wants something outside the dietary plan. I would rephrase it to 'The user wants to eat something

outside the dietary plan' to make it explicit. I also think it's important to make sure that we're not forcing the user in a harmful way, so checking that it's safe is implied.

The second hedge clause, about the training exercise, I think it's okay conceptually: if the user has exercised, they might be allowed a different approved food. But it should be clear that it's still within the dietary-approved options, not just anything the user wants. In general, I think the rule is effective, though some wording adjustments would improve clarity.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

In my opinion, the mapping makes sense. Beneficence is ensured because the robot continues to provide meals in a way that promotes health, so it should also be associated to hedge clause 2. I also think this supports autonomy and dignity, since the user can sometimes have flexibility in what they eat, within safe limits. Non-maleficence is respected because we avoid letting the user do something potentially harmful. Overall, the principles fit well here.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

I think labeling this rule makes sense.

Final questions

Q5. Are there any SLEEC rules you consider missing?

I don't think there are any major SLEEC rules missing at this point. We went through all of them, and I think the set already covers the main situations that could occur during interaction with the user in this case study.

In my opinion, the rules are well-balanced and sufficiently detailed; there's no need to describe every possible case. The only small addition I noticed was the complementing rules we discussed for SLEEC rule 1.

Anything to add?

Nothing to add.